



Tehran University of Medical Sciences

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Final Exam
International Campus, MD Program
Tissue Development & Function
September 2019 semester

Name and Surname:

Student ID Number:

Course Coordinator: Dr. Rastegar

Date of Exam: Jan 27, 2020

Duration of Exam: 60 Minutes

Key

MCQ Question	5
Short Answer Questions	6

Directions:

- Each of the following questions contains one correct answer only.
- Please choose the correct answer for the MCQs in the answer sheet with pencil.
- For short answer Questions, write down your answer in the question pamphlet



Physiology: Dr: Ranjbaran

1. Which of the following keeps the myosin and actin filaments in place?

- a. Titin
- b. Z discs
- c. Dense bodies
- d. Calsequestrin

2. Which of the following is the main cause of rigor mortis?

- a. Ca ions release in the sarcoplasm
- b. Release of ADP and P from the myosin filaments
- c. Absence of ATP
- d. High ATPase activity of myosin heads

3. Which of the following inhibits actin filaments in the skeletal muscle fibers?

- a. Curare
- b. Botulinum toxin
- c. Tropomyosin troponin complex
- d. Ca ions

4. Suppose that in a smooth muscle, myosin kinase and myosin phosphatase are both strongly activated. What happens in this condition?

- a. Contraction cycle stops in relaxed state.
- b. Cycling frequency of myosin heads is great.
- c. Little energy is used by the muscle.
- d. Long-term maintenance of tonic contraction is seen.

5. Which of the following is not matched with excitation contraction coupling in the skeletal muscles?

- a. Action potential
- b. Ryanodine receptors
- c. Multiple adherent points
- d. Transverse tubules

6. In which muscle, control is exerted mainly by nerve signals?

- a. Ureters
- b. Bile Duct
- c. Ciliary Muscle
- d. Uterus



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7. Which of the following is the possible cause of the staircase effect?

- a. Progressive increase of calcium ions in the sarcoplasm
- b. Frequency Summation
- c. Multiple Fiber Summation
- d. Tetanization

8. What is the main function of Ca – calmodulin dependent protein kinase in the skeletal muscle neuro-muscular junction?

- a. Phosphorylation of myosin light chain kinase
- b. Activation of myosin phosphatase
- c. Phosphorylation of synapsin
- d. Muscle contraction

9. Which of the following best describes the importance of stress relaxation phenomenon?

- a. Maintaining of the same amount of volume inside the lumen
- b. Increase in pressure in response to increase in volume
- c. Maintaining of the same amount of pressure inside the lumen
- d. Increase in pressure in response to prolonged tonic contraction

10. In an electromyography report, findings are compatible with severe axon lesion of a nerve supply to a skeletal muscle. What symptom do you expect to see in this patient?

- a. Hyperplasia of muscle fibers
- b. Atrophic changes
- c. Muscle fatigue
- d. Remodeling of muscle fibers

11. Regulatory protein in iris muscle is

- a. troponin complex
- b. tropomyosin
- c. calmodulin
- d. myosin

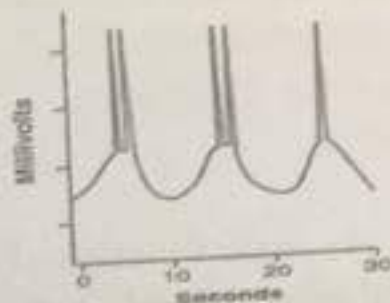
12. Caveolae is the rudimentary analog of the

- a. transverse tubule system
- b. sarcoplasmic reticulum
- c. dense bodies
- d. Z discs

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13. In which muscle is this type of action potential seen?

- a. Skeletal muscle
- b. Cardiac muscle
- c. Multi-unit smooth muscle
- d. Unitary smooth muscle



Histology: Dr. Rastegar

14. Which layer of the epidermis is present only in thick skin?

- A. Stratum basale
- B. Stratum spinosum
- C. Stratum lucidum
- D. Stratum corneum

15. Which cell is a macrophage found in the skin?

- A. Langerhans cell
- B. Dust cell
- C. Microglia
- D. Histiocyte

16. What type of glands are the ceruminous glands in ear?

- A. Sebaceous glands
- B. Eccrine gland
- C. Endocrine gland
- D. Apocrine gland

Histology: Dr. Minaee

17. Which of the following is Observed in central nervous System?

- A - Schwann Cell
- B - Remack Cell
- C - Microglial Cell
- D - Noruglia

18. Which of the following Histological Structure Is Observed Bipolar cell?

- A - inner Ear
- B - Ganglion
- C - central nervous System
- D - nothing

19. In which part of Nervous system Is Observed myelinated nerve fiber With Oligodendrocyte?

- A - peripheral nervous System
- B - inner Ganglion
- C - outer Ganglion
- D - central nervous System



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20. Which of the following is Observed in peripheral nervous System ?

A – Ganglions

B – Ependymal Cells

C – Cerebral Spinal Fluid

D- Nothing

Histology: Dr. Malek

21. Russel bodies are characteristic for which cell?

a) mast cell

b) plasmacell

c) kupfer

d) Adipocyte

22. structure of framework of spleen and lymphnodes are composed of which fibers?

a) reticular

b) elastic

c) elastin

d) collagen bundles

23. which one is dense irregular connective tissue?

a) Dermis of skin

c) Adipose tissue

b) Tendon

d) Areolar tissue

24. Red blood cell is because it has In the wall.

a) fragile-actin

c) flexible-spectrin

b) flexible-fibrinogen

d) fragile-spectrin

25. which cell has most number in the normal blood smear?

a) Neutrophil

b) Basophil

c) monocyte

d) Eosinophil

26. in which cell we have major basic protein?

a). lymphocyte

c) macrophage

b) monocyte

d) Eosinophil

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Embryology: Dr. Amidi

27-Somites

- A. are the source of cells in the nucleus pulposus
- B. give rise to cardiac muscle
- C. give rise to the cartilages of the branchial arches
- D. are derived from the paraxial mesoderm

28-Maternal serum alpha-fetoprotein increases if the fetus has

- A. oligohydramnios
- B. Bochdalek hernia
- C. anencephaly
- D. Meningocele

29-Implantation of the blastocyst cannot occur without prior

- A. differentiation of syncytiotrophoblast
- B. hatching from the zona pellucida
- C. fragmentation of the primary yolk sac
- D. development of amnioblast

30-Syncytiotrophoblast

- A. secretes chorionic gonadotropin
- B. gives rise to the decidua basalis
- C. begins to develop on the 4th day postfertilization
- D. is derived from the zona pellucida

31-Cells in somites give rise to

- A. dorsal root ganglia
- B. the mandible
- C. vertebrae
- D. cardiac muscle

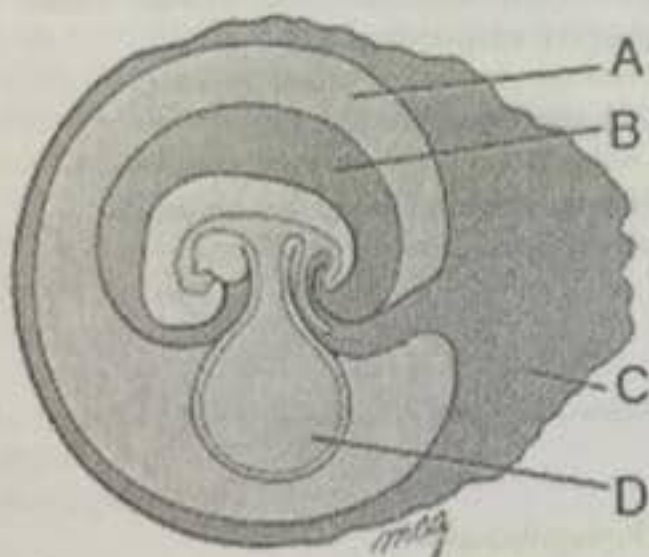
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32-Concerning the placenta

- A.it is fed by fetal arteries that travel along the secondary yolk sac
- B.it is derived primarily from amnioblast and allantois
- C.it is comprised of villous chorion and decidua basalis**
- D.it begins to secrete chorionic gonadotropin at the end of the first month

33-36

Questions ~~34~~ refer to the image below. For each of these questions, choose the **SINGLE BEST** answer.



33. The approximate age of this embryo is:

- A.1 week
- B.2 weeks
- C.about 4 weeks**
- D.about 12 weeks

34. The label indicating the amniotic cavity is:

- A.A
- B.B**
- C.C
- D.D

35. The label indicating a space lined by endoderm is:

- A.A
- B.B
- C.C
- D.D**

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36. The label indicating a space containing fluid that is produced by the kidneys in the developing fetus is

- A.A
- B.B**
- C.C
- D.D

37. The placenta barrier that exists throughout most of pregnancy is composed of.....

- A. endothelium and connective tissue
- B. Endothelium and cytotrophoblast
- C. Connective tissue and cytotrophoblast
- D. syncytiotrophoblast and endothelium**

38. The extraembryonic coelom is..... the cavity

- A. Amniotic
- B. Yolk sac
- C. Allantoic
- D. Chorionic**

39. Following fertilization of the ovum, the ovarian corpus luteum continues to produce progesterone. If the corpus luteum regresses in the first trimester, abortion occurs. Which of the following hormones maintains this capability of the corpus luteum?

- a. Alpha-fetoprotein
- b. Estradiol
- c. Human chorionic gonadotropin**
- d. Human placental lactogen

40. Which of the following disorders is associated with a defect in ciliary dynein and infertility?

- a. Cystic fibrosis
- b. Kennedy syndrome
- c. Kartagener's syndrome**
- d. X-linked situs inversus abnormalities

41. The inner cell mass is also called

- a. Morula
- b. Gastrula
- c. Embryoblast**
- d. Nerula

42. Notochordal cells remain to form which of the following structures in the newborn?

- a. Vertebral arches
- b. Vertebral bodies
- c. Annulus fibrosus
- d. Nucleus pulposus**

43. Embryonic hematopoiesis occurs first in the

- a. Primary stem villi
- b. Primary yolk sac
- c. Secondary stem villi
- d. Secondary yolk sac**

44. Which category: Non random concurrence of two or more features that occur more frequently than expected by chance alone but for which no etiology has been demonstrated

- a. Sequence
- b. Spectrum**
- c. Complex
- d. Association



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45-The biopsy of the ovary of an 8 year old female reveals the presence of a normal ovary the cortex of which consists predominantly of which type of follicle?

- a. Corpus luteum
- b. Primary follicles

- c. Primordial follicles
- d. Mature follicles

46-An abnormal shape or position of a part of the body that results from mechanical forces is called

- a. Dysplasia
- b. Malformation

- c. Disruption
- d. Deformation

47-The intraembryonic mesoderm is formed by

- a. Some cells of the primitive streak displacing hypoblast
- b. Some mesenchymal tissue

- c. Epiblast
- d. All of above

48-Which statements regarding primary oocytes are correct?

- a. Primary oocytes begin at first meiotic division after birth
- b. primary oocytes form after birth
- c. secondary oocytes complet befor ovulation
- d. Completion of prophase miosis I occurs after LH surge

49-The part of the chorion that helps form the placenta is called

- a) Chorion laevae
- b) Chorion villi

- c) Chorion frondosum
- d) Chorion cavity

50-In which of the following infections Usually child infected by mother at birth?

- a. Rubella
- b. Cytomegalovirus

- c. Herpes simplex
- d. Toxoplasmosis

51-In which of the following conditions would sex chromatin (Barr bodies) be absent

- a. 46 (XX)
- b. 46 (XY)

- c. 47 (XXY)
- d. 47 (XX)

52-The lens placode is formed by which one of the following embryonic tissues

- a. Neural ectoderm
- b. Neural crest cells

- c. Surface ectoderm
- d. Somatic mesoderm

53-Oligohydramnios is associated with

- a) Atresia of esophagus
- b) Renal agenesis

- c) Atresia of duodenum
- d) Atresia of stomach